

2023-Ph-H1-Q19

Section: Particles and Waves

Topic: Interference

Two coherent sound sources produce a loud sound. What causes this?

Answer: A

(Waves arrive in phase)

Guidance for Students:

In-phase waves interfere **constructively**, reinforcing each other and increasing amplitude → loud sound.

Out of phase → destructive interference → silence.

Revision Tips:

- In-phase = crest + crest (or trough + trough).
- Phase difference = 0 or multiples of 2π
- Coherence: constant phase relationship.